# Predicting NIMBYism Causally

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## 1. What problem am I solving? State it clearly.

Investigating causal drivers of neighborhood opposition (NIMBYism) to housing [Einstein et al., 2019, p. 28]. **Goal:** separate stable, **invariant** predictors of resistance from **spurious associations** driven by **environment-specific confounders** [Peters et al., 2018, p. 84]. **Aim:** find features predicting opposition consistently across the distinct **market regimes** of the 2018–2025 period.

## 2. Why is this problem important? To whom?

Critical for urban planners. Current models overfit to specific eras, failing when market regimes shift [Arjovsky et al., 2019, p. 1]. Identifying *invariant* causes enables policy design addressing fundamental concerns rather than symptoms of a moment, ensuring **robustness** [Bühlmann, 2020, p. 404].

## 3. What is my strategy for solving it?

Using **Invariant Risk Minimization (IRM)** [Arjovsky et al., 2019, p. 5] and **Bayesian Invariant Prediction (BIP)** [Wu et al., 2025b, p. 2], both grounded in the seminal **Invariant Prediction** framework [Peters et al., 2016, p. 947]:

1. **Environments:** Partition data via **expanding window Cross-Validation** into **Training Environments** ($\mathcal{E}_{tr}$) and **Test Environments** ($\mathcal{E}_{te}$):

| **Window** | **Training ($\mathcal{E}_{tr}$)** | **Test ($\mathcal{E}_{te}$)** |
|---|---|---|
| 1 | 2018 | 2019–2025 |
| 2 | 2018–2019 | 2020–2025 |
| ... | ... | ... |
| $N$ | 2018–2024 | 2025 |

2. **Representation:** Extract features $\Phi(X)$ from property data and protest letters [Schölkopf et al., 2021, p. 15].
3. **Methods:** We implement the two core paradigms scheduled for **Weeks 6–7** of the seminar [Blei, 2026]:
   - **Invariant Risk Minimization (IRM):** A *regularized optimization* paradigm to learn $\Phi$ such that a **fixed dummy classifier** $w = 1.0$ is simultaneously **optimal** across all $\mathcal{E}_{tr}$, **enforced** via a **gradient norm penalty** [Arjovsky et al., 2019, p. 5]:

$$\min_{\Phi} \sum_{e \in \mathcal{E}_{tr}} R^e(\Phi) + \lambda \|\nabla_{w|w=1.0} R^e(w \cdot \Phi)\|^2$$

   - **Bayesian Invariant Prediction (BIP):** A *probabilistic inference* paradigm defining a latent **feature selection vector** $z \in \{0,1\}^p$; the **posterior** $p(z|D)$ **targets the true invariant features** [Wu et al., 2025b, p. 2].

## 4. Why is this solution good? Where does it fall short?

**It falls short because** it assumes an **invariant causal mechanism** [Peters et al., 2018, p. 107]. The invariance assumption fails if the underlying reasons for NIMBYism undergo **mechanism**

**change** over time [Peters et al., 2018, p. 107]. For instance, neighborhood opposition may shift from traffic congestion fears to displacement concerns as demographics evolve.

## 5. How can I demonstrate that the solution works?

Train on expanding $\mathcal{E}_{tr}$, test on all **future horizons** (2019–2025) in $\mathcal{E}_{te}$. Successful Invariant Model demonstrates **predictive stability** through the **invariance of its AUC** score across all horizons, maintaining reliability where baselines fail under **distribution shift** [Krueger et al., 2021, p. 1].

## References


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